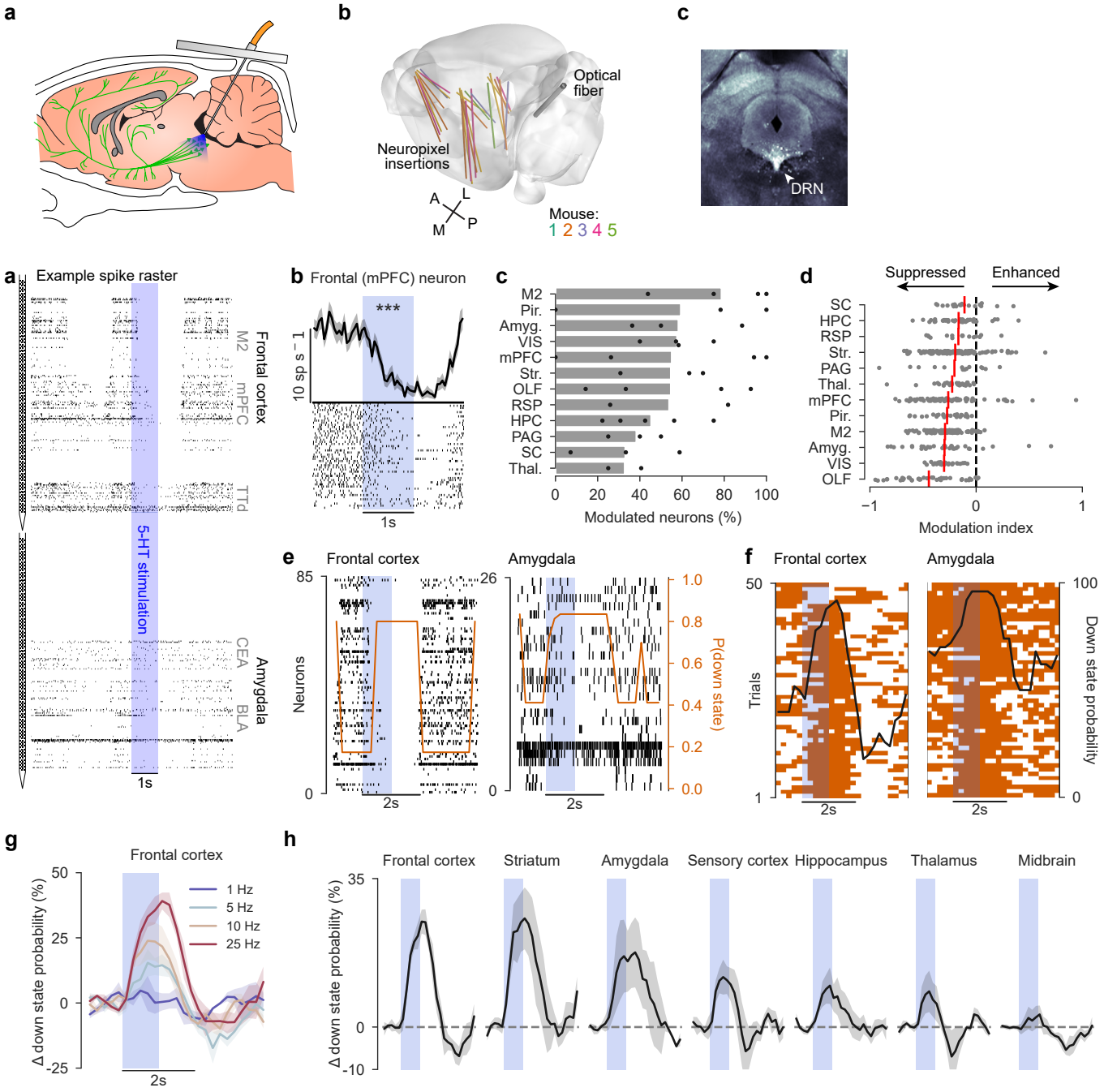

a
b
c
Optical fiber
Neuropixel insertions
A
L
M
P
Mouse:
1 2 3 4 5
DRN
a
Example spike raster
Frontal cortex
M2
mPFC
TTd
5-HT stimulation
Amygdala
CEA
BLA
1s
b
Frontal (mPFC) neuron
***
10 sp s − 1
1s
c
M2
Pir.
Amyg.
VIS
mPFC
Str.
OLF
RSP
HPC
PAG
SC
Thal.
0
20
40
60
80
100
Modulated neurons (%)
d
Suppressed
Enhanced
SC
HPC
RSP
Str.
PAG
Thal.
mPFC
Pir.
M2
Amyg.
VIS
OLF
−1
0
1
Modulation index
e
Frontal cortex
Amygdala
85
0
Neurons
26
0
2s
1.0
0.8
0.6
0.4
0.2
0.0
P(down state)
f
Frontal cortex
Amygdala
50
1
Trials
100
0
Down state probability
2s
g
Frontal cortex
50
25
0
-25
Δ down state probability (%)
1 Hz
5 Hz
10 Hz
25 Hz
2s
h
Frontal cortex
Striatum
Amygdala
Sensory cortex
Hippocampus
Thalamus
Midbrain
35
0
-10
Δ down state probability (%)